Mini Project is a **Digital Clock System using FPGA (Verilog HDL)**. The project requires designing separate modules and verifying them using an **Integrated Logic Analyzer (ILA)**.

Project Title

FPGA-Based Digital Clock Using Verilog HDL

Aim

To design and implement a digital clock on an FPGA that counts **seconds, minutes, and hours** using sequential logic. The system uses a **100 MHz input clock**, which is divided into **1 Hz** before driving the counters.

Block Diagram Explanation

The project consists of the following modules:

1. **27-bit Counter (Clock Divider)**
    - Input: 100 MHz clock
    - Output: Slow clock (approximately 1 Hz)
2. **Bit Selector**
    - Selects the required counter bit to generate the 1 Hz clock.
3. **Seconds Counter**
    - Counts from **0 to 59**.
    - After reaching 59, it resets to 0 and increments the minute counter.
4. **Minutes Counter**
    - Counts from **0 to 59**.
    - After reaching 59, it resets and increments the hour counter.
5. **Hours Counter**
    - Counts from **0 to 23**.
    - After reaching 23, it resets to 0.
6. **Integrated Logic Analyzer (ILA)**
    - Used to observe the outputs of the seconds, minutes, and hours counters inside the FPGA.

Objectives

- Design each block as a separate Verilog module.
- Generate a 1 Hz clock from a 100 MHz FPGA clock.
- Implement seconds, minutes, and hours counters.
- Connect all modules together.
- Verify the design using the Integrated Logic Analyzer (ILA).

Required Modules

- `clock_divider.v`
- `bit_selector.v`
- `seconds_counter.v`
- `minutes_counter.v`
- `hours_counter.v`
- `top_module.v`

Working Principle

1. FPGA receives a **100 MHz clock**.
2. The clock divider reduces it to **1 Hz**.
3. Every 1-second pulse increments the seconds counter.
4. When seconds reach 59:
   - Seconds reset to 0.
   - Minutes increase by 1.
5. When minutes reach 59:
   - Minutes reset.
   - Hours increase by 1.
6. When hours reach 23:
   - Hours reset to 0.
7. The ILA monitors all counter outputs.

Expected Output

- Seconds: **00–59**
- Minutes: **00–59**
- Hours: **00–23**
- Continuous 24-hour digital clock operation.

Tools Required

- Vivado Design Suite
- Verilog HDL
- FPGA Board (NIELIT/Xilinx)
- Integrated Logic Analyzer (ILA)
